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Review

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Immunopathology of synovitis: from histology to molecular pathways

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Abstract

Increased knowledge about pathological processes active in inflammatory joint diseases is needed to initiate personalized medicine based on targeted treatments in the future. The molecular and cellular pathways that are active during joint inflammation may differ between the various inflammatory joint diseases, between different patient subgroups within one disease, or even between different stages of the disease in a single patient. In this review, we evaluate synovial inflammation in terms of descriptive histopathology through to more functional studies on human synovial tissue inflammation in RA and SpA, in phenotypic subgroups of RA and SpA patients, and during the disease course of both diseases.

Key words: rheumatoid arthritis, histopathology, spondyloarthritis, synovium, inflammation, microarrays/gene expression profiling/transcriptome, spondylarthropathies, cytokines and inflammatory mediators, fibroblasts, lymphocytes, macrophages.

REVIEW

Rheumatology key messages

- Pathogenic cellular and molecular pathways differ between various synovial diseases.
- Pathogenic pathways differ between RA and spondyloarthritis patient subgroups and different phases of disease.
- Disease outcome in RA and spondyloarthritis depends on the histological and molecular context.

Introduction

Personalized medicine based on targeted treatments in the individual patient is one of the goals set for future healthcare. In order to achieve this goal in rheumatology we have to increase our knowledge of the pathological cellular and molecular processes that initiate, drive and perpetuate joint inflammation. These pathophysiologic processes may differ between the various inflammatory joint diseases, between different patient subgroups within one disease, or even between different stages of the disease in a single patient. The synovium is one of the major target tissues of RA and SpA, the two most prevalent types of chronic inflammatory joint diseases, and the availability of synovial tissue collected in different disease phases provides a unique opportunity to study the cellular and molecular pathways driving chronic joint inflammation. In this review, we describe the characteristics of synovial tissue immunopathology in a range of inflammatory joint diseases. Second, we compare synovial inflammatory changes within the phenotypic subgroups of RA and SpA. Third, we discuss the cellular and molecular features of synovitis during the disease course, including pre- and post-treatment (Fig. 1). Finally, we indicate important perspectives for future research.

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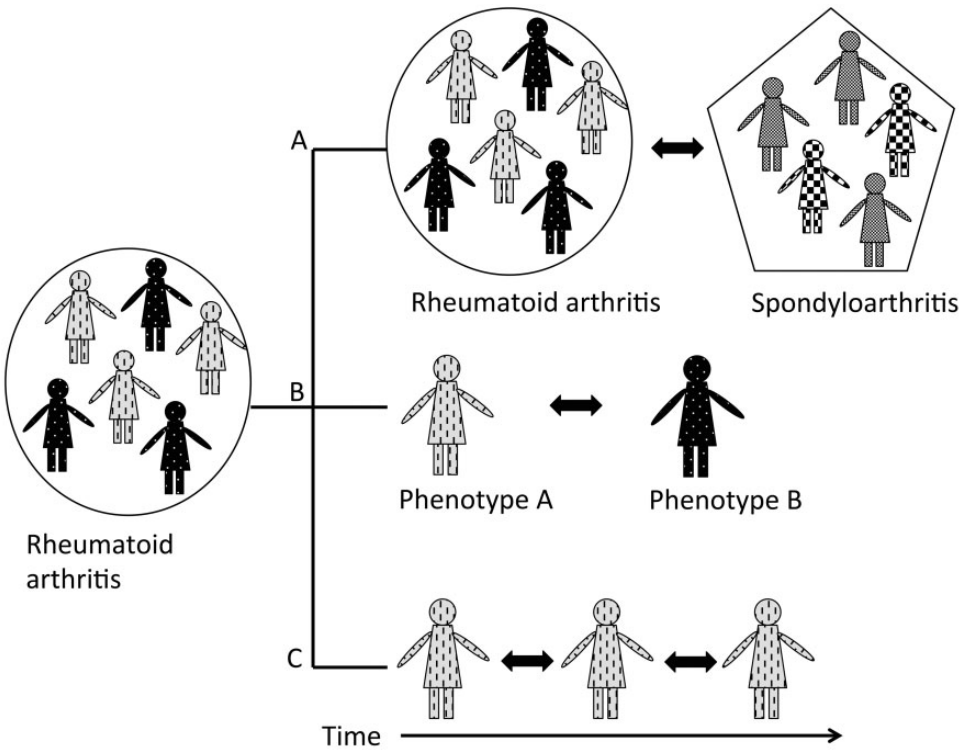
Synovial tissue in health and disease

The synovium is a membrane attached to skeletal tissue at the bone–cartilage interface that borders the joint cavities and that lines tendon sheaths and bursae. A healthy synovium consists of a thin, 1–2-cell-thick layer, intimal lining layer of fibroblast-like synoviocytes and macrophages, and a sublining layer of loose connective tissue with blood vessels, lymph vessels, fibroblasts, collagen fibres, nerve fibres and only very few leucocytes.

Inflamed synovial tissue displays three major histological alterations: hyperplasia of the intimal lining layer due to accumulation of macrophages and proliferation of fibroblast-like synoviocytes that are in an altered activation state; neoangiogenesis with endothelial activation in the synovial sublining; and a vast influx of inflammatory cells such as macrophages, dendritic cells, lymphocytes

Fig. 1 Research in three different dimensions



(**A**) Between different diseases. (**B**) Between different phenotypical subtypes of disease. (**C**) Serial analysis in the same patient (group).

and mast cells in the synovial sublining. These infiltrating leucocytes are activated and produce a vast amount of pro-inflammatory and destructive mediators that contribute to synovitis as well as to cartilage and bone destruction.

In contrast to synovial tissue obtained during joint replacement therapy in end-stage disease, several less invasive techniques to obtain synovial tissue have been developed [1, 2]. This allows studying the cellular and molecular characteristics of the inflamed synovium in full-blown active disease, in early arthritis, or even in the pre-clinical phase of the disease [3, 4]. Moreover, these techniques allow sequential sampling of the same joint over time in a single patient, which enables evaluation of treatment effects with anti-inflammatory and targeted therapies.

Blind needle biopsy, mini-arthroscopy and US-guided synovial tissue biopsy sampling have been standardized for research purposes [5]. Synovial tissue sampling can be performed safely in both large joints (knee, ankle) and small joints (e.g. wrist and MCP joints). Although mini-arthroscopy is a somewhat invasive procedure, patients tolerate it very well. It has been shown that inflammation in one joint is similar to inflammation in other joints [6]. A limited number of 6–8 biopsies taken from various locations in the joint accurately reflects the spatial heterogeneous inflammatory infiltrate [7] and gene expression levels [8] within the joint. Synovial tissue macrophages and IL-6 levels have been shown to correspond well to local knee pain [9]. Both in SpA and in RA, changes in CD68-positive sublining macrophages correlate well with disease activity and response to treatment [10–13]. This shows that changes seen in synovial tissue analysis truly reflect the inflammatory process.

Comparative synovial immunohistology between arthritic diseases

Several studies have compared the synovial immunopathological features between various inflammatory joint diseases. Most of these studies investigated RA *vs* SpA. Only a few studies included other inflammatory joint diseases such as gout, OA or JIA [14–19]. The hypothesis of these studies was that, in addition to many non-specific inflammatory features, synovitis may display disease-specific cellular and molecular alterations that supposedly reflect key pathogenic events and which, if highly specific, could even be applied as diagnostic markers.

Histological characterization of synovitis in RA *vs* SpA

Of the three major histological hallmarks of synovial inflammation—hyperplasia of the synovial lining layer, neoangiogenesis and infiltration of the synovial sublining with inflammatory cells—only vascularization appears to be consistently different between RA and SpA. Both in early and established disease, vascularization is increased in SpA compared with RA [14, 20–22]. Also, molecules involved in angiogenesis are differently expressed in RA compared with in SpA [23]. The angiopoietin-1/Tie2 axis was shown to be specifically upregulated in early RA, even before ACR EULAR criteria for RA are met, whereas in SpA increased expression of Angiopoietin-2 was present [24].

Although the global degree of synovial infiltration is not different between RA and SpA, more detailed characterization of the infiltrating cells by immunohistochemistry revealed distinct features. Synovial infiltration by mast cells, neutrophils and CD163 macrophages is increased in SpA compared with in RA [14, 22, 25–28] In contrast, the numbers of fibroblast-like synoviocytes, CD68-positive macrophages, T cells, B cells and plasma cells have been shown to be similar in RA and SpA, both in early and established disease [15, 16, 22, 29]. It has been suggested that high numbers of macrophages and plasma cells could predict a diagnosis of RA in recent-onset arthritis patients [15], but this finding could not be confirmed in larger, independent patient cohorts [30, 31].

Not only the number and type of infiltrating leucocytes, but also their spatial organization in the synovial micro-architecture may be of pathophysiological relevance. Three types of lymphocyte infiltration patterns are observed in synovitis: diffuse infiltration of lymphocytes, aggregates of B cells and T cells, and ectopic lymphoid neogenesis characterized by lymphocyte aggregates with prototypical features of germinal centres, such as the presence of follicular dendritic cells and high endothelial venules [32]. Because the latter histologically resemble secondary lymphoid tissues, it was proposed that synovial ectopic lymphoid neogenesis may play a role in mounting immune responses, and in particular the autoimmune responses observed in RA [33]. However, a series of studies demonstrated that lymphocyte aggregates and ectopic lymphoid neogenesis are equally found in RA and SpA, are not associated with specific autoimmune features (such as the presence of RA-specific autoantibodies), and are a reflection of the degree of local tissue inflammation [16, 34–36].

Molecular characterization of synovitis in RA *vs* SpA

Besides quantification and phenotypic characterization of resident and infiltrating cell populations in synovitis, novel more sophisticated techniques allowed studying in more detail the specific molecular mechanisms that are active within the cells and that determine cellular function. These molecular processes may be differentially regulated between inflammatory joint diseases, even when the cellular infiltrate is comparable as shown by immunohistochemistry.

From a pathophysiologic point of view, RA is considered an autoimmune disease in which the self-reactive lymphocytes play an important role. This is reflected by synovial tissue local autoantibody production [37] and features characteristic of antigen-specific lymphocyte activation. For instance, the human cartilage glycoprotein-39 was identified as a candidate T cell auto-antigen in RA, and the immunodominant epitope of this autoantigen is presented by the RA-associated HLA-DR4 molecule in the inflamed rheumatoid joint [38, 39]. Using a unique antibody selectively recognizing these MHC-peptide complexes, we were able to demonstrate that this molecular feature is highly specific for RA synovitis compared with SpA synovitis [26, 40]. As to humoral autoimmunity in RA, the most specific autoantibody family consists of autoantibodies against citrullinated proteins. Although a variety of citrullinated proteins, and in particular citrullinated fibrinogen, are found in all types of synovial inflammation [41], several studies have reported that intracellular citrullinated proteins were specifically found in RA synovitis [42, 43]. Both features—HC gp-39/MHC complexes and intracellular citrullinated presence—appeared to be highly specific for RA synovitis and were demonstrated to be valuable diagnostic biomarkers in a large prospective study [30].

In contrast to RA, SpA is regarded as an autoinflammatory disease with an important role for the innate immune system (reviewed in [44, 45]). This is reflected by the increase in CD163-positive macrophages, neutrophils and mast cells in SpA *vs* RA synovitis [25, 27]. None of these features is specific enough to be used as a diagnostic biomarker, but they appear to reveal important pathophysiological pathways. For example, mast cells are not only increased in SpA but also express more IL-17 than mast cells in RA, and targeting mast cells via a c-Kit inhibitor resulted in decreased levels of IL17 *ex vivo* [25]. Anti-IL17 treatment has shown clinical effect in a proof of concept study in AS, the prototype of SpA [46, 47].

A second example is the more detailed molecular analysis of stromal cells in SpA *vs* RA. Although the number of synovial fibroblasts and their morphology on histology seems quite similar between the two diseases, a pangenomic gene expression analysis showed a robust, highly significant and reproducible upregulation of myogenes in SpA [37]. Additional immunohistochemical staining showed that myofibroblasts specifically express these myogenes. The myofibroblast signature was independent of disease duration, treatment and clinical phenotype of SpA and, most interestingly, was not modulated by clinically and biologically effective downmodulation of synovial inflammation by TNF blockade. Further research on the functional role of these cells in SpA is needed.

Collectively, the cellular and molecular characterization of synovitis in RA *vs* SpA supports a prominent role for (autoimmune) T and B cells in RA *vs* innate immune cells and stromal cells in SpA. Accordingly, treatment strategies that target innate immune cells and their mediators might be most effective in SpA, whereas in RA, B- or T-cell–targeted therapies are more successful.

It is important to note that routine histology and characterization of the cellular infiltrate does not allow discrimination between the major forms of arthritis, and therefore synovial tissue biopsy sampling is rarely used in clinical care as a diagnostic tool. In some cases, where the synovial fluid analysis doesn't show abnormalities, urate or calcium pyrophosphate crystals can be detected in synovial tissue biopsies. Diagnosis of chronic bacterial infection (tuberculosis, Lyme, fungal infection), sarcoidosis, leukaemic infiltration, pigmented villonodular synovitis, reticulohistocytosis, amyloidosis, haemochromatosis and ochronosis might require synovial tissue histology [2]. Although the use of more detailed molecular tools, as discussed above, does allow better diagnostic discrimination, these tools still require further validation in larger prospective cohorts.

Comparative synovial histopathology within arthritic diseases

Both RA and SpA are clinically heterogeneous diseases. Traditionally both diseases have been subdivided into several phenotypic subtypes. In RA a distinction between seropositive and seronegative RA, and between erosive and non-erosive RA is made. SpA includes AS, PsA, reactive arthritis and IBD-related arthritis. Patients with these subtypes of SpA can suffer from both peripheral and/or axial symptoms. The question arises whether the clinical features that are in part overlapping and/or contrasting between the disease subtypes represent one underlying pathogenic entity with a somewhat different clinical phenotype or, on the contrary, that there are several different underlying molecular mechanisms that share a common clinical phenotype.

RA

In RA, ACPA-positive RA is thought to be a molecularly/pathophysiologically distinct subtype. ACPA-positive RA is associated with the shared epitope and smoking [48]. Most studies comparing ACPA-positive and ACPA-negative RA did not show any differences in the synovial cellular infiltrate between the two subgroups [16, 35]. Only one study showed increased lymphocyte infiltration in ACPA-positive RA compared with ACPA-negative RA, which possibly reflects the increased systemic disease activity, because the ACPA-positive patients had a higher clinical disease activity [11, 49].

A study on T-cell repertoire in ACPA-positive RA, ACPA-negative RA and SpA showed a restricted T cell repertoire specifically in ACPA-positive RA. This was independent of numbers of infiltrating CD3-positive T cells, as these were similar in both groups. Moreover, this restricted repertoire was unique for the synovial tissue, as it could not be shown in peripheral blood T cells [50]. The exact phenotype, origin, function and (auto)antigen specificity of these clones remains to be elucidated.

As to erosive *vs* non-erosive RA, a few cross-sectional studies suggested some differences, such as in expression of P53 [51], metalloproteinase 2 expression [52, 53] and human cartilage gp-39 [54]. Data on lymphocyte infiltration patterns showed conflicting results [34, 36, 55]. The issue with the cross-sectional design, however, is that the findings may be biased by selection of the patients. Prospective studies in early arthritis that bypass these limitations have until now not been able to detect synovial markers with enough power to predict development of erosive disease in the individual patient. Studies on larger cohorts of early arthritis patients are awaited, not only to correlate specific markers to the development of erosive disease but also to increase insight into the functional property of these markers in relation to the development of erosions.

In contrast to studies that compare subgroups defined beforehand, gene expression analysis by microarray allows unbiased clustering of patients based on the gene expression profiles that are present. In RA, the patients' three subtypes of gene expression profiles were identified by microarray analysis of synovial tissue biopsies. The first subtype was characterized by genes involved in the adaptive immune response, *MMP 1* and *MMP 3* genes, STAT-encoding and -induced genes and antigen-presenting-cell–related genes. The second subtype was characterized by genes involved in extracellular matrix remodelling. The last subtype was comparable with a gene expression profile in OA [56]. A study on gene expression profiles in fibroblast-like synoviocytes (FLSs) identified two types of FLS. High-inflammatory FLSs were characterized by expression of a TGF-β/activin A–inducible gene profile and FLSs from low inflammatory synovial tissue showed increased expression of growth factor genes [57]. However, the relevance of these findings remains unclear because these molecular subtypes, as defined by gene expression profiling, were not demonstrated to correlate with clinically relevant outcomes, such as disease activity, structural damage or response to treatment. Collectively these data support the existence of a range of subtypes of synovitis within RA, but large-scale prospective studies are needed to delineate the pathophysiological and/or clinical relevance of these subtypes.

SpA

Both axial and peripheral joint involvement can be seen in SpA. It remains very challenging to obtain tissue from axial SpA, and no good head-to-head comparative studies have been performed to evaluate inflamed tissue characteristic differences between axial and peripheral SpA. In peripheral SpA it has been shown that AS, undifferentiated SpA and PsA show similar synovial infiltrate, characterized by increased levels of CD163 macrophages, neutrophils and mast cells [21, 28]. Also, the increased vascularity and the myogene signature are shared between the various peripheral SpA subtypes, arguing that they represent one pathophysiological entity [58].

SpA does not only affect axial and peripheral joints, but is also characterized by extra-articular manifestations, such as skin psoriasis and IBD [59]. Comparison of synovial immunopathology with skin or gut shows some interesting similarities, but also significant differences. In the context of the gut–joint axis, for example, an increase in $CD163^+$ macrophages is not only seen in the joints of SpA patients but also in their gut, mimicking the findings in

Crohn's disease [27, 60]. Similarly, both skin and joint show increased vascularization, T cells and TNFα expression in PsA. Nevertheless, functional interpretation of these data requires caution, because T cell-directed therapies such as alefacept appear to be very effective for psoriasis but not for PsA, suggesting that different molecular pathways are active in the skin and the joint.

Comparative synovial histopathology during the disease course of arthritic diseases

Natural history of RA

Besides the heterogeneity within one disease, there might also be a variety of pathogenic mechanisms active during the disease course from initiation of joint inflammation to persistent chronic inflammation in a single patient. Dynamic sampling of the joint over time, either in cohorts or even better in prospective studies, allows comparison of so-called early RA (often defined as <1 year of symptoms) with established RA. When controlling for disease activity and treatment, the cellular infiltrate and the expression of inflammatory molecules, such as cytokines, chemokines, adhesion molecules, metalloproteinases and granzymes, is comparable between early and long-standing (>5-year disease duration) RA [9, 14, 61, 62], suggesting that so-called early RA already depicts chronic synovitis. This topic, however, was recently reassessed using more detailed molecular approaches. Using state-of-the-art deep sequencing technology [63], molecular studies on T-cell repertoire in early and established RA showed increased number of highly expanded T-cell clones in early RA [64]. If these clones represent autoreactive T cells responding to a local autoantigen, this warrants further research. Similarly, expanded B cell clones were specifically observed in the synovial tissue of early arthritis patients [65]. These studies teach us two lessons: first, a variety of cellular and/or molecular pathways are indeed activated during the disease course of RA and, second, we need sophisticated molecular technologies to trace the activation of these pathways.

An exciting new development is the study of the synovial tissue in the pre-clinical phase of RA, as we still do not know exactly when and how synovial disease is initiated. The presence of RA-specific auto-antibodies in the serum of healthy individuals several years before they develop clinically overt RA [66, 67] has enabled studies on the disease in the pre-clinical phase. In individuals with increased serum levels of ACPA and/or IgM rheumatoid factor, inflammatory cells and vasculature in the synovial tissue were not increased compared with in healthy controls [3, 4]. A trend towards an increased number of T cells being related to the development of arthritis was seen in autoantibody-positive individuals compared with healthy controls [4]. As T cells are thought to play an important role in RA development, further studies focusing on T-cell function in this pre-clinical phase might show clear differences between individuals who develop or do not develop arthritis over time.

Treatment effects

Another interesting application of dynamic synovial histopathology over time is the assessment of treatment effects. Synovial biopsies taken before and after initiation of treatment can reveal the biological effects of these treatments. The change in CD68-positive macrophages, for example, is an established biomarker for evaluating treatment response in RA and SpA, because changes in synovial macrophages correlate well with clinical response [10-13, 68]. This biomarker can thus help in evaluating potential treatment effects in proof-of-concept trials with new compounds.

In addition to evaluating treatment response, synovial tissue studies before and after treatment can help us to understand the mechanism of action of a specific drug. Studies on TNFα blockade have shown that the decrease in synovial tissue inflammatory cells after treatment is not the result of apoptosis [69], but might be a result of the egress of cells via lymphatic vessels. Studies in RA patients and mice have shown an upregulation of lymphatic vessels after treatment with infliximab [70].

It is not only important to evaluate the biological effects of a treatment, but also to predict whether a treatment will be effective or not in an individual patient. Indeed, an increasing number of very effective targeted therapies have become available for the treatment of RA and other rheumatic diseases, but the response to all these drugs is quite heterogeneous between patients. It becomes increasingly important to be able to predict who will or will not respond to a specific therapeutic agent in order to prevent exposure of patients to the potential side effects of an ineffective drug and to lower costs for ineffective treatments. Studies before and after treatment have been used to try to identify specific markers that predict treatment response. In RA patients treated with infliximab, the baseline synovial tissue expression of TNFα correlated with clinical response to infliximab [36]. Contrarily, microarray could not predict response to infliximab in RA [71]. For rituximab treatment, it has been shown that gene expression profiles [72] and baseline synovial tissue interferon gene signature [73] are associated with treatment response to rituximab in RA. Unfortunately, none of these findings have been confirmed in independent replication studies, and the degree of correlation of specific features with treatment response was too weak to predict response in individual patients.

The future: crossing the border of the human synovium

Although much of the histopathological research in rheumatic diseases has focused on the synovium as the primary target organ of the disease, emerging insights suggest that the diseases may be initiated in other organ systems and, accordingly, that other tissues should be histologically and molecularly evaluated. One example is autoantibody-positive RA, in which the original autoimmune trigger may originate in the lung, the lymph nodes or even the peridonteum [74]. A couple of histopathological studies have started to assess these organs [75, 76], but more research in this direction is warranted. The same holds true for SpA, where early events could occur in the gut, the enthesis or the bone before affecting the synovial tissue.

A second important focus of future research should be to correlate the findings in human synovium with *in vivo* observations in animal models. For example, the relationship between joint inflammation and new bone formation in SpA is very difficult to study in human patients, because osteoproliferation occurs mainly in axial joints, which are difficult to sample, and because progression of osteoproliferation takes many years. Therefore, the use of animal models such as the HLA-B27 tg rats can be very useful in complementing the analysis of human target tissues [77].

Finally, our analysis of arthritic tissue should not remain restricted to classical immunopathological approaches. As demonstrated by the microarray discovery of the myogene signature in SpA synovitis [58] and the deep-sequencing analysis of T and B cell clonality in the RA synovium [64, 65], we need to foster the use of state-of-the-art molecular technologies in the analysis of synovial immunopathology.

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References

1 Baeten D, Van Den Boseche F, Elewaut D *et al*. Needle arthroscopy of the knee with synovial biopsy sampling: technical experience in 150 patients. Clin Rheumatol 1999;18:434–41.

2 Gerlag DM, Tak PP. How to perform and analyse synovial biopsies. Best Pract Res Clin Rheumatol 2009;23:221–32.

3 van de Sande MG, de Hair MJ, van der Leij C *et al*. Different stages of rheumatoid arthritis: features of the synovium in the preclinical phase. Ann Rheum Dis 2011;70:772–7.

4 de Hair MJ, van de Sande MG, Ramwadhdoebe TH *et al*. Features of the synovium of individuals at risk of developing rheumatoid arthritis: implications for understanding preclinical rheumatoid arthritis. Arthritis Rheumatol 2014;66:513–22.

5 van de Sande MG, Gerlag DM, Lodde BM *et al*. Evaluating antirheumatic treatments using synovial biopsy: a recommendation for standardisation to be used in clinical trials. Ann Rheum Dis 2011;70:423–7.

6 Kraan MC, Reece RJ, Smeets TJ *et al*. Comparison of synovial tissues from the knee joints and the small joints of rheumatoid arthritis patients: implications for pathogenesis and evaluation of treatment. Arthritis Rheum 2002;46:2034–8.

7 Dolhain RJ, Ter Haar NT, de Kuiper R *et al*. Distribution of T cells and signs of T-cell activation in the rheumatoid joint: implications for semiquantitative comparative histology. Br J Rheumatol 1998;37:324–30.

8 Boyle DL, Rosengren S, Bugbee W, Kavanaugh A, Firestein GS. Quantitative biomarker analysis of synovial gene expression by real-time PCR. Arthritis Res Ther 2003;5:R352–60.

9 Tak PP, Smeets TJ, Daha MR *et al*. Analysis of the synovial cell infiltrate in early rheumatoid synovial tissue in relation to local disease activity. Arthritis Rheum 1997;40:217–25.

10 Haringman JJ, Gerlag DM, Zwinderman AH *et al*. Synovial tissue macrophages: a sensitive biomarker for response to treatment in patients with rheumatoid arthritis. Ann Rheum Dis 2005;64:834–8.

11 Wijbrandts CA. Absence of changes in the number of synovial sublining macrophages after ineffective treatment for rheumatoid arthritis: implications for the use of synovial sublining macrophages as a biomarker. Arthritis Rheum 2007;56:3869–71.

12 Kruithof E, De Rycke L, Vandooren B *et al*. Identification of synovial biomarkers of response to experimental treatment in early-phase clinical trials in spondylarthritis. Arthritis Rheum 2006;54:1795–804.

13 Baeten D, Houbiers J, Kruithof E *et al*. Synovial inflammation does not change in the absence of effective treatment: implications for the use of synovial histopathology as biomarker in early phase clinical trials in rheumatoid arthritis. Ann Rheum Dis 2006;65:990–7.

14 Baeten D, Demetter P, Cuvelier C *et al*. Comparative study of the synovial histology in rheumatoid arthritis, spondyloarthropathy, and osteoarthritis: influence of disease duration and activity. Ann Rheum Dis 2000;59:945–53.

15 Kraan MC, Haringman JJ, Post WJ *et al*. Immunohistological analysis of synovial tissue for differential diagnosis in early arthritis. Rheumatology 1999;38:1074–80.

16 van de Sande MG, Thurlings RM, Boumans MJ *et al*. Presence of lymphocyte aggregates in the synovium of patients with early arthritis in relationship to diagnosis and outcome: is it a constant feature over time? Ann Rheum Dis 2011;70:700–3.

17 Da RR, Qin Y, Baeten D, Zhang Y. B cell clonal expansion and somatic hypermutation of Ig variable heavy chain genes in the synovial membrane of patients with osteoarthritis. J Immunol 2007;178:557–65.

18 Canete JD, Celis R, Noordenbos T *et al*. Distinct synovial immunopathology in Behcet disease and psoriatic arthritis. Arthritis Res Therap 2009;11:R17.

19 Kruithof E, Van den Bossche V, De Rycke L *et al*. Distinct synovial immunopathologic characteristics of juvenile-onset spondylarthritis and other forms of juvenile idiopathic arthritis. Arthritis Rheum 2006;54:2594–604.

20 Fraser A, Fearon U, Reece R, Emery P, Veale DJ. Matrix metalloproteinase 9, apoptosis, and vascular morphology in early arthritis. Arthritis Rheum 2001;44:2024–8.

21 Kruithof E, Baeten D, De Rycke L *et al*. Synovial histopathology of psoriatic arthritis, both oligo- and polyarticular, resembles spondyloarthropathy more than it does rheumatoid arthritis. Arthritis Res Ther 2005;7:R569–80.

22 Baeten D, Kruithof E, De Rycke L *et al*. Infiltration of the synovial membrane with macrophage subsets and polymorphonuclear cells reflects global disease activity in spondyloarthropathy. Arthritis Res Ther 2005;7:R359–69.

23 Fearon U, Griosios K, Fraser A *et al*. Angiopoietins, growth factors, and vascular morphology in early arthritis. J Rheumatol 2003;30:260–8.

24 van de Sande MG, de Launay D, de Hair MJ *et al*. Local synovial engagement of angiogenic TIE-2 is associated with the development of persistent erosive rheumatoid arthritis in patients with early arthritis. Arthritis Rheum 2013;65:3073–83.

25 Noordenbos T, Yeremenko N, Gofita I *et al*. Interleukin-17-positive mast cells contribute to synovial inflammation in spondylarthritis. Arthritis Rheum 2012;64:99–109.

26 Baeten D, Steenbakkers PG, Rijnders AM *et al*. Detection of major histocompatibility complex/human cartilage gp-39 complexes in rheumatoid arthritis synovitis as a specific and independent histologic marker. Arthritis Rheum 2004;50:444–51.

27 Baeten D, Demetter P, Cuvelier CA *et al*. Macrophages expressing the scavenger receptor CD163: a link between immune alterations of the gut and synovial inflammation in spondyloarthropathy. J Pathol 2002;196:343–50.

28 De Rycke L, Baeten D, Foell D *et al*. Differential expression and response to anti-TNFalpha treatment of infiltrating versus resident tissue macrophage subsets in autoimmune arthritis. J Pathol 2005;206:17–27.

29 van Kuijk AW, Reinders-Blankert P, Smeets TJ, Dijkmans BA, Tak PP. Detailed analysis of the cell infiltrate and the expression of mediators of synovial inflammation and joint destruction in the synovium of patients with psoriatic arthritis: implications for treatment. Ann Rheum Dis 2006;65:1551–7.

30 Baeten D, Kruithof E, De Rycke L *et al*. Diagnostic classification of spondylarthropathy and rheumatoid arthritis by synovial histopathology: a prospective study in 154 consecutive patients. Arthritis Rheum 2004;50:2931–41.

31 van de Sande MG, de Hair MJ, Schuller Y *et al*. The features of the synovium in early rheumatoid arthritis according to the 2010 ACR/EULAR classification criteria. PLoS ONE 2012;7:e36668.

32 Manzo A, Bugatti S, Caporali R *et al*. CCL21 expression pattern of human secondary lymphoid organ stroma is conserved in inflammatory lesions with lymphoid neogenesis. Am J Pathol 2007;171:1549–62.

33 Humby F, Bombardieri M, Manzo A *et al*. Ectopic lymphoid structures support ongoing production of class-switched autoantibodies in rheumatoid synovium. PLoS Med 2009;6:e1.

34 Canete JD, Celis R, Moll C *et al*. Clinical significance of synovial lymphoid neogenesis and its reversal after anti-tumour necrosis factor alpha therapy in rheumatoid arthritis. Ann Rheum Dis 2009;68:751–6.

35 Cantaert T, Kolln J, Timmer T *et al*. B lymphocyte autoimmunity in rheumatoid synovitis is independent of ectopic lymphoid neogenesis. J Immunol 2008;181: 785–94.

36 Thurlings RM, Wijbrandts CA, Mebius RE *et al*. Synovial lymphoid neogenesis does not define a specific clinical rheumatoid arthritis phenotype. Arthritis Rheum 2008;58:1582–9.

37 Masson-Bessiere C, Sebbag M, Durieux JJ *et al*. In the rheumatoid pannus, anti-filaggrin autoantibodies are produced by local plasma cells and constitute a higher proportion of IgG than in synovial fluid and serum. Clin Exp Immunol 2000;119:544–52.

38 van Lierop MJ, den Hoed L, Houbiers J *et al*. Endogenous HLA-DR-restricted presentation of the cartilage antigens human cartilage gp-39 and melanoma inhibitory activity in the inflamed rheumatoid joint. Arthritis Rheum 2007;56:2150–9.

39 Tran CN, Davis MJ, Tesmer LA *et al*. Presentation of arthritogenic peptide to antigen-specific T cells by fibroblast-like synoviocytes. Arthritis Rheum 2007;56: 1497–506.

40 Steenbakkers PG, Baeten D, Rovers E *et al*. Localization of MHC class II/human cartilage glycoprotein-39 complexes in synovia of rheumatoid arthritis patients using complex-specific monoclonal antibodies. J Immunol 2003;170:5719–27.

41 Chapuy-Regaud S, Sebbag M, Baeten D *et al*. Fibrin deimination in synovial tissue is not specific for rheumatoid arthritis but commonly occurs during synovitides. J Immunol 2005;174:5057–64.

42 Baeten D, Peene I, Union A *et al*. Specific presence of intracellular citrullinated proteins in rheumatoid arthritis synovium: relevance to antifilaggrin autoantibodies. Arthritis Rheum 2001;44:2255–62.

43 De Rycke L, Nicholas AP, Cantaert T *et al*. Synovial intracellular citrullinated proteins colocalizing with peptidyl arginine deiminase as pathophysiologically relevant antigenic determinants of rheumatoid arthritis-specific humoral autoimmunity. Arthritis Rheum 2005;52:2323–30.

44 Ambarus C, Yeremenko N, Tak PP, Baeten D. Pathogenesis of spondyloarthritis: autoimmune or autoinflammatory? Curr Opin Rheumatol 2012;24:351–8.

45 Dougados M, Baeten D. Spondyloarthritis. Lancet 2011;377:2127–37.

46 Baeten D, Baraliakos X, Braun J *et al*. Anti-interleukin-17A monoclonal antibody secukinumab in treatment of ankylosing spondylitis: a randomised, double-blind, placebo-controlled trial. Lancet 2013;382:1705–13.

47 Baeten DL, Kuchroo VK. How Cytokine networks fuel inflammation: interleukin-17 and a tale of two autoimmune diseases. Nat Med 2013;19:824–5.

48 Klareskog L, Stolt P, Lundberg K *et al*. A new model for an etiology of rheumatoid arthritis: smoking may trigger HLA-DR (shared epitope)-restricted immune reactions to autoantigens modified by citrullination. Arthritis Rheum 2006;54:38–46.

49 van Oosterhout M, Bajema I, Levarht EW *et al*. Differences in synovial tissue infiltrates between anti-cyclic citrullinated peptide-positive rheumatoid arthritis and anti-cyclic citrullinated peptide-negative rheumatoid arthritis. Arthritis Rheum 2008;58:53–60.

50 Cantaert T, Brouard S, Thurlings RM *et al*. Alterations of the synovial T cell repertoire in anti-citrullinated protein antibody-positive rheumatoid arthritis. Arthritis Rheum 2009;60:1944–56.

51 Salvador G, Sanmarti R, Garcia-Peiro A *et al*. p53 expression in rheumatoid and psoriatic arthritis synovial tissue and association with joint damage. Ann Rheum Dis 2005;64:183–7.

52 Goldbach-Mansky R, Lee JM, Hoxworth JM *et al*. Active synovial matrix metalloproteinase-2 is associated with radiographic erosions in patients with early synovitis. Arthritis Res 2000;2:145–53.

53 Kraan MC, Haringman JJ, Weedon H *et al*. T cells, fibroblast-like synoviocytes, and granzyme B+ cytotoxic cells are associated with joint damage in patients with recent onset rheumatoid arthritis. Ann Rheum Dis 2004;63:483–8.

54 Baeten D, Boots AM, Steenbakkers PG *et al*. Human cartilage gp-39+,CD16+ monocytes in peripheral blood and synovium: correlation with joint destruction in rheumatoid arthritis. Arthritis Rheum 2000;43:1233–43.

55 Klimiuk PA, Goronzy JJ, Bjor NJ, Beckenbaugh RD, Weyand CM. Tissue cytokine patterns distinguish variants of rheumatoid synovitis. Am J Pathol 1997;151:1311–9.

56 van der Pouw Kraan TC, van Gaalen FA, Huizinga TW *et al*. Discovery of distinctive gene expression profiles in rheumatoid synovium using cDNA microarray technology: evidence for the existence of multiple pathways of tissue destruction and repair. Genes Immun 2003;4:187–96.

57 Kasperkovitz PV, Verbeet NL, Smeets TJ *et al*. Activation of the STAT1 pathway in rheumatoid arthritis. Ann Rheum Dis 2004;63:233–9.

58 Yeremenko N, Noordenbos T, Cantaert T *et al*. Disease-specific and inflammation-independent stromal alterations in spondylarthritis synovitis. Arthritis Rheum 2013;65:174–85.

59 Rudwaleit M, Baeten D. Ankylosing spondylitis and bowel disease. Best Pract Res Clin Rheumatol 2006;20:451–71.

60 Demetter P, De Vos M, Van Huysse JA *et al*. Colon mucosa of patients both with spondyloarthritis and Crohn’s disease is enriched with macrophages expressing the scavenger receptor CD163. Ann Rheum Dis 2005;64:321–4.

61 Tak PP, Thurkow EW, Daha MR *et al*. Expression of adhesion molecules in early rheumatoid synovial tissue. Clin Immunol Immunopathol 1995;77:236–42.

62 Smeets TJ, Dolhain RJ, Breedveld FC, Tak PP. Analysis of the cellular infiltrates and expression of cytokines in synovial tissue from patients with rheumatoid arthritis and reactive arthritis. J Pathol 1998;186:75–81.

63 Klarenbeek PL, Tak PP, van Schaik BD *et al*. Human T-cell memory consists mainly of unexpanded clones. Immunol Lett 2010;133:42–8.

64 Klarenbeek PL, de Hair MJ, Doorenspleet ME *et al*. Inflamed target tissue provides a specific niche for highly expanded T-cell clones in early human autoimmune disease. Ann Rheum Dis 2012;71:1088–93.

65 Doorenspleet ME, Klarenbeek PL, de Hair MJ *et al*. Rheumatoid arthritis synovial tissue harbours dominant B-cell and plasma-cell clones associated with autoreactivity. Ann Rheum Dis 2014;73:756–62.

66 Nielen MM, van Schaardenburg D, Reesink HW *et al*. Specific autoantibodies precede the symptoms of rheumatoid arthritis: a study of serial measurements in blood donors. Arthritis Rheum 2004;50:380–6.

67 Rantapaa-Dahlqvist S, de Jong BA, Berglin E *et al*. Antibodies against cyclic citrullinated peptide and IgA rheumatoid factor predict the development of rheumatoid arthritis. Arthritis Rheum 2003;48:2741–9.

68 Bresnihan B, Gerlag DM, Rooney T *et al*. Synovial macrophages as a biomarker of response to therapeutic intervention in rheumatoid arthritis: standardization and consistency across centers. J Rheumatol 2007;34:620–2.

69 Wijbrandts CA, Remans PH, Klarenbeek PL *et al*. Analysis of apoptosis in peripheral blood and synovial tissue very early after initiation of infliximab treatment in rheumatoid arthritis patients. Arthritis Rheum 2008;58:3330–9.

70 Polzer K, Baeten D, Soleiman A *et al*. Tumour necrosis factor blockade increases lymphangiogenesis in murine and human arthritic joints. Ann Rheum Dis 2008;67:1610–6.

71 Lindberg J, Wijbrandts CA, van Baarsen LG *et al*. The gene expression profile in the synovium as a predictor of the clinical response to infliximab treatment in rheumatoid arthritis. PLoS One 2010;5:e11310.

72 Hogan VE, Holweg CT, Choy DF *et al*. Pretreatment synovial transcriptional profile is associated with early and late clinical response in rheumatoid arthritis patients treated with rituximab. Ann Rheum Dis 2012;71:1888–94.

73 Thurlings RM, Boumans M, Tekstra J *et al*. Relationship between the type I interferon signature and the response to rituximab in rheumatoid arthritis patients. Arthritis Rheum 2010;62:3607–14.

74 Mikuls TR, Payne JB, Yu F *et al*. Periodontitis and *Porphyromonas gingivalis* in patients with rheumatoid arthritis. Arthritis Rheumatol 2014;66:1090–100.

75 Makrygiannakis D, Hermansson M, Ulfgren AK *et al*. Smoking increases peptidylarginine deiminase 2 enzyme expression in human lungs and increases citrullination in BAL cells. Ann Rheum Dis 2008;67:1488–92.

76 de Hair MJ, Zijlstra IA, Boumans MJ *et al*. Hunting for the pathogenesis of rheumatoid arthritis: core-needle biopsy of inguinal lymph nodes as a new research tool. Ann Rheum Dis 2012;71:1911–2.

77 van Duivenvoorde LM, Dorris ML, Satumtira N *et al*. Relationship between inflammation, bone destruction, and osteoproliferation in the HLA-B27/human beta2-microglobulin-transgenic rat model of spondylarthritis. Arthritis Rheum 2012;64:3210–9.